

JINPENG ZHANG

Software Engineer · Backend / Frontend / Fullstack · MSc Student at Aalborg University
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EDUCATION

•Aalborg University

MSc in Software Engineering

2025 – Present

Copenhagen, Denmark

•Ca' Foscari University of Venice

BSc in Computer Science

2020 – 2024

Venice, Italy

PROFESSIONAL EXPERIENCE

•Themison

Backend Developer Intern (AI / RAG)

Nov 2025 - Feb 2026

Copenhagen, Denmark

- Engineered the retrieval pipeline of a production **RAG** system, benchmarking embedding models and chunking strategies to improve semantic search accuracy and improved **Chain-of-Thought** prompting, reducing LLM hallucinations.

•e-project s.r.l.

Programmer Analyst (Fullstack)

Jul 2024 – Aug 2025

Padua, Italy

- Analyzed client requirements from detailed specifications, tracked them in Excel, and translated them into database changes via custom scripts and stored procedures.
- Developed and tested database solutions in **Oracle SQL Developer**, cross-validated outcomes on the **AS400 / IBM i** platform, and documented progress and results in detailed technical reports.
- Implemented new features for a banking web application as a fullstack engineer: **Spring Boot**, **JHipster**, and **Liquibase** on the backend with **Angular** on the frontend, delivering fully integrated solutions and supporting new microservices.

•GPI Group

Java Developer – Full SDLC

Sep 2023 – Jul 2024

Mestre, Italy

- Trained in data and report management on **AS400 / IBM i** systems using the **RPGLE** language; defined and created databases for application data with focus on optimal structure and performance.
- Developed and implemented backend services and business functionality in **Spring Boot**, ensuring seamless operation and scalability of the healthcare platform.
- Led the **full lifecycle development** of a new module in **Java 8** – requirements, design, **Java Swing** UI implementation, **JUnit** testing, **Maven** build, deployment, and integration into the existing software.

ENGINEERING PROJECTS

•ARIADNE – Indoor Navigation Platform

Python / FastAPI, Neo4j, PostGIS, Docker Compose, Swift / SwiftUI, YOLO, ONNX, WebSockets, RAG

github.com/Jimpoz

- Architected a multi-service indoor navigation platform with a **FastAPI** gateway/middleware, a spatial backend, an LLM **RAG** assistant, and two computer-vision services, all orchestrated via **Docker Compose**.
- Designed a hybrid storage layer combining **Neo4j** (graph of truth, GDS Dijkstra pathfinding) mirrored to **Supabase PostGIS** (GiST-indexed geometry) through **SQLAlchemy** + **GeoAlchemy2**.
- Built real-time AR way-finding by streaming JPEG frames over **WebSocket** to a **YOLOv2 ONNX** detector, plus a batch **YOLOv11** pipeline producing structured room summaries from uploaded photos.
- Shipped the **iOS** client (**SwiftUI**, Combine, AVFoundation, CoreLocation) with floor-plan rendering, live camera detection overlay, and an LLM-powered "Ask for Directions" chat grounded in the graph.

•Information Origin Tracker – Distributed NLP Infrastructure

Kubernetes, Spring Boot, PostgreSQL, NVIDIA Container Toolkit, Qwen LLM

- Orchestrated a self-hosted **Kubernetes** cluster (WSL2) running GPU-accelerated LLM inference (**Qwen**) via the **NVIDIA Container Toolkit**.
- Designed and integrated LLM-based **NLP** pipelines for contextual summarization and top-K sentence retrieval.
- Built a **Spring Boot** middleware for persistence in PostgreSQL and secured remote access using **Ngrok** tunneling and **Tailscale** mesh networking.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript/JavaScript, Swift, SQL (Cypher), RPGLE, C/C++

Backend: Spring Boot, JHipster, FastAPI, Starlette, REST APIs, WebSockets, Microservices, JUnit, Maven

Frontend / UI: Angular, React, SwiftUI, Combine, Java Swing, HTML5, CSS3, Tailwind, responsive design

AI & Data: RAG, LangChain, Hugging Face, PyTorch, sentence-transformers, YOLO (Ultralytics, ONNX), pgvector, Pandas, NLTK, Prompt Engineering (CoT)

Cloud & DevOps: Docker, Docker Compose, Kubernetes, Linux (WSL2), Git, CI/CD, NVIDIA Container Toolkit, Ngrok, Tailscale

Databases: PostgreSQL, PostGIS, Neo4j (Cypher, GDS), Oracle (SQL Developer), DB2, Supabase, SQLite